

CHAPTER – XII

APPENDIX

12.1 – Source Code

App.py

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def home():
    return render_template('index.html')

@app.route('/about')
def about():
    return render_template('about.html')

@app.route('/dashboard01')
def dashboard01():
    return render_template('dashboard01.html')

@app.route('/dashboard02')
def dashboard02():
    return render_template('dashboard02.html')

@app.route('/dashboard03')
def dashboard03():
    return render_template('dashboard03.html')

@app.route('/story')
def story():
    return render_template('story.html')

if __name__ == '__main__':
    app.run(debug=True)
```

Base.html

```
<!DOCTYPE html>
<html lang="en">

  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
      scale=1.0">

    <title>
      {% block title %}Plugging into the Future{% endblock %}
    </title>

    <!-- Bootstrap -->
    <link
      href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.
        min.css"
      rel="stylesheet">

    <!-- Custom CSS -->
    <link rel="stylesheet"
      href="{{ url_for('static', filename='css/style.css') }}">
    </head>

    <body>

      {% include 'navbar.html' %}

      <main>
        <!-- {% block content %}-->
        <!-- {% endblock %}-->
      </main>

      <footer class="bg-dark text-white text-center py-3 mt-5">
        <p class="mb-0">
          © 2026 Plugging into the Future | Electricity Consumption
          Analytics
        </p>
      </footer>

      <script
        src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bu
          ndle.min.js"></script>
```

```
</body>
</html>
```

Index.html

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Plugging into the Future</title>

    <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.
min.css" rel="stylesheet">

    <style>

      body{
font-family: Arial, sans-serif;
background:#f8f9fa;
      }

      /* Navbar */
      .navbar{
background:#0d1b2a;
      }

      .navbar-brand{
color:white !important;
font-weight:bold;
      }

      .nav-link{
color:white !important;
      }

      .nav-link:hover{
color:#fffc107 !important;
      }
```

```
/* Hero */
.hero{
  text-align:center;
  padding:60px 20px;
}

.hero h1{
  font-size:4rem;
  font-weight:700;
  color:#23395d;
}

.hero p{
  font-size:1.4rem;
  color:#555;
}

.dashboard-btn{
  margin-top:20px;
  padding:12px 30px;
  font-size:20px;
}

/* Highlights */
.section-title{
  text-align:center;
  margin-top:50px;
  margin-bottom:30px;
  color:#23395d;
  font-weight:bold;
}

.highlight-card{
  background:white;
  border-radius:8px;
  box-shadow:0 2px 8px rgba(0,0,0,0.15);
  padding:25px;
  text-align:center;
  transition:0.3s;
}

.highlight-card:hover{
```

```
transform:translateY(-5px);
    }

    .highlight-card h4{
        color:#23395d;
        font-weight:bold;
    }

    footer{
        margin-top:50px;
        padding:20px;
        text-align:center;
        color:#666;
    }

</style>
</head>

<body>

    <!-- Navbar -->
<nav class="navbar navbar-expand-lg navbar-dark">
    <div class="container">

        <a class="navbar-brand" href="#">
            ⚡ Plugging into the Future
        </a>

        <button class="navbar-toggler"
            type="button"
            data-bs-toggle="collapse"
            data-bs-target="#navMenu">
            <span class="navbar-toggler-icon"></span>
        </button>

        <div class="collapse navbar-collapse" id="navMenu">

            <ul class="navbar-nav ms-auto">

                <li class="nav-item">
                    <a class="nav-link" href="">Home</a>
                </li>
```

```
<li class="nav-item">
  <a class="nav-link" href="/about">About</a>
</li>
```

```
<li class="nav-item">
  <a class="nav-link" href="/dashboard01">Dashboard 1</a>
</li>
```

```
<li class="nav-item">
  <a class="nav-link" href="/dashboard02">Dashboard 2</a>
</li>
```

```
<li class="nav-item">
  <a class="nav-link" href="/dashboard03">Dashboard 3</a>
</li>
```

```
<li class="nav-item">
  <a class="nav-link" href="/story">Story</a>
</li>
```

```
</ul>
```

```
</div>
```

```
</div>
```

```
</nav>
```

```
<!-- Hero Section -->
```

```
<section class="hero">
```

```
<div class="container">
```

```
<h1>
```

```
  Plugging into the Future:<br>
  An Exploration of Electricity Consumption Patterns
</h1>
```

```
<p>
```

```
  Interactive Data Visualization Using Tableau
```

```
</p>
```

```
<a href="/dashboard01"
class="btn btn-success dashboard-btn">
  View Dashboard
</a>

</div>

</section>

<!-- Highlights -->
<div class="container">

  <h2 class="section-title">
    Project Highlights
  </h2>

  <div class="row">

    <div class="col-md-4 mb-3">
      <div class="highlight-card">
        <h4>Consumption Analysis</h4>
        <p>
          Study electricity usage trends across different regions.
        </p>
      </div>
    </div>

    <div class="col-md-4 mb-3">
      <div class="highlight-card">
        <h4>Regional Comparison</h4>
        <p>
          Compare energy demand among states and sectors.
        </p>
      </div>
    </div>

    <div class="col-md-4 mb-3">
      <div class="highlight-card">
        <h4>Interactive Dashboard</h4>
        <p>
          Explore insights using Tableau visualizations.
        </p>
      </div>
    </div>
  </div>
</div>
```



```
</div>
</div>

</div>

</div>

<!-- About -->
<section id="about" class="container mt-5">

  <h2 class="text-center mb-4">
    About the Project
  </h2>

  <p class="text-center">
    This project analyzes electricity consumption patterns,
    identifies trends, compares regional demand, and provides
    future energy forecasts through interactive dashboards.
  </p>

</section>

<footer>
  © 2026 Plugging into the Future: An Exploration of Electricity
  Consumption Patterns
</footer>

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></script>

</body>
</html>
```

About.html

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
```



```
<title>About Project</title>
```

```
    <link  
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.  
min.css" rel="stylesheet">
```

```
<style>
```

```
    body{  
        background:#f4f7fc;  
        font-family:'Segoe UI',sans-serif;  
    }
```

```
    /* Project Overview */  
    .overview-box{  
background:linear-gradient(135deg,#0d6efd,#6610f2);  
        color:white;  
        border-radius:20px;  
        padding:50px;  
        box-shadow:0 10px 30px rgba(0,0,0,0.15);  
    }
```

```
    /* Team Section */  
    .team-section{  
        padding:70px 0;  
    }
```

```
        .team-title{  
            text-align:center;  
            font-weight:bold;  
            color:#23395d;  
            margin-bottom:50px;  
        }
```

```
        .team-card{  
            background:white;  
            border:none;  
            border-radius:20px;  
            overflow:hidden;  
            box-shadow:0 5px 20px rgba(0,0,0,0.12);  
            transition:all 0.3s ease;  
            text-align:center;
```

```
height:100%;  
}
```

```
.team-card:hover{  
  transform:translateY(-10px);  
  box-shadow:0 12px 25px rgba(0,0,0,0.18);  
}
```

```
.profile-circle{  
  width:100px;  
  height:100px;  
  background:#0d6efd;  
  color:white;  
  border-radius:50%;  
  display:flex;  
  align-items:center;  
  justify-content:center;  
  font-size:36px;  
  font-weight:bold;  
  margin:25px auto 15px;  
}
```

```
.team-card h5{  
  font-weight:bold;  
  color:#23395d;  
}
```

```
.role{  
  font-weight:bold;  
  margin-bottom:15px;  
}
```

```
.leader{  
  color:#0d6efd;  
}
```

```
.analyst{  
  color:#198754;  
}
```

```
.developer{  
  color:#fd7e14;
```

```
}

.researcher{
  color:#dc3545;
}

.card-text{
  color:#666;
  padding:0 15px 20px;
}

/* Footer */
.footer{
  background:#23395d;
  color:white;
  text-align:center;
  padding:20px;
  margin-top:40px;
}

</style>

</head>

<body>

<!-- Project Overview -->
<section class="container py-5">

<div class="overview-box text-center">

  <h1 class="fw-bold">
    ⚡ Plugging into the Future
  </h1>

  <h3 class="mb-4">
    An Exploration of Electricity Consumption Patterns
  </h3>

  <hr class="w-25 mx-auto">

  <p class="lead">
```

Plugging into the Future: An Exploration of Electricity Consumption Patterns

This project explores electricity consumption patterns using advanced data analytics, visualization techniques, and predictive modeling.

</p>

<p>

The objective is to identify trends in energy consumption, compare usage across regions and sectors, and generate insights that support efficient energy management and future planning.

</p>

</div>

</section>

<!-- Team Section -->

<section class="container team-section">

<h2 class="team-title">

Meet Our Team

</h2>

<div class="row g-4">

<!-- Team Member 1 -->

<div class="col-lg-3 col-md-6">

<div class="team-card">

<div class="profile-circle">

SO

</div>

<div class="card-body">

<h5>SRINIVASAN OMPRAKASH</h5>

<p class="role leader">

Team Leader

</p>

<p class="card-text">

Responsible for project planning,
team coordination, implementation,
testing and final documentation.

</p>

</div>

</div>

</div>

<!-- Team Member 2 -->

<div class="col-lg-3 col-md-6">

<div class="team-card">

<div class="profile-circle">

VS

</div>

<div class="card-body">

<h5>V SANJAY</h5>

<p class="role analyst">

Data Analyst

</p>

<p class="card-text">

Performed data collection,
data cleaning, processing,
analysis and visualization.

</p>

</div>

</div>

</div>

<!-- Team Member 3 -->

<div class="col-lg-3 col-md-6">

<div class="team-card">

<div class="profile-circle">

JP

</div>

```
<div class="card-body">
```

```
<h5>JANGAM PURUSHOTHAM</h5>
```

```
<p class="role developer">  
  Dashboard Developer  
</p>
```

```
<p class="card-text">  
  Designed interactive dashboards,  
  developed UI components and  
  implemented visual analytics.  
</p>
```

```
</div>  
</div>  
</div>
```

```
<!-- Team Member 4 -->
```

```
<div class="col-lg-3 col-md-6">  
  <div class="team-card">
```

```
    <div class="profile-circle">  
      DF  
    </div>
```

```
<div class="card-body">
```

```
<h5>DUDEKULA MOHAMMAD FARUK</h5>
```

```
<p class="role researcher">  
  Research Specialist  
</p>
```

```
<p class="card-text">  
  Conducted literature review,  
  research analysis and prepared  
  project reports and findings.  
</p>
```

```
</div>
```

```
</div>
</div>

</div>

</section>

<!-- Footer -->
<div class="footer">
<h5>⚡ Plugging into the Future</h5>
<p class="mb-0">
An Exploration of Electricity Consumption Patterns
</p>
</div>

</body>
</html>
```

DASHBOARD-1.html

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Dashboard 01</title>

    <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.
min.css" rel="stylesheet">

    <style>
      html, body {
        height: 100%;
        margin: 0;
        overflow: hidden; /* Prevent page scrolling */
      }

      .container-fluid {
        height: 100vh;
        padding: 10px;
      }
```



```
.card {
  height: calc(100vh - 20px);
}

.card-body {
  height: 100%;
  display: flex;
  flex-direction: column;
}

.tableauPlaceholder {
  flex: 1;
  width: 100%;
}
</style>
</head>
<body>

<div class="container-fluid">

  <div class="card shadow h-100">
    <div class="card-body">

      <h4 class="text-center mb-3">
        Monthly Electricity Consumption
      </h4>

      <div class="tableauPlaceholder"
        id="viz1781513166776"
        style="position: relative;">

        <noscript>
          <a href="#">
            
          </a>
        </noscript>

      <object class="tableauViz" style="display: none;">
```

```
        <param name="host_url"
value="https%3A%2F%2Fpublic.tableau.com%2F"/>
        <param name="embed_code_version" value="3"/>
        <param name="path" value="shared/8FXG62QGR"/>
        <param name="toolbar" value="yes"/>
        <param name="static_image"
value="https://public.tableau.com/static/images/8F/8FXG62QGR/1.png"/>
        <param name="animate_transition" value="yes"/>
        <param name="display_static_image" value="yes"/>
        <param name="display_spinner" value="yes"/>
        <param name="display_overlay" value="yes"/>
        <param name="display_count" value="yes"/>
        <param name="language" value="en-US"/>
        <param name="filter" value="publish=yes"/>
    </object>
</div>
</div>
</div>
</div>
<script>
var divElement = document.getElementById('viz1781513166776');
var vizElement = divElement.getElementsByTagName('object')[0];

vizElement.style.width = '100%';
vizElement.style.height = '100%';

var scriptElement = document.createElement('script');
scriptElement.src =
'https://public.tableau.com/javascripts/api/viz_v1.js';
vizElement.parentNode.insertBefore(scriptElement, vizElement);
</script>

</body>
</html>
```

DASHBOARD-2.html

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Dashboard 02</title>

    <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.
min.css" rel="stylesheet">

    <style>
      *{
        margin:0;
        padding:0;
        box-sizing:border-box;
      }

      html,
      body{
        width:100%;
        height:100%;
        overflow:hidden;
      }

      .container-fluid{
        width:100vw;
        height:100vh;
        padding:0;
        margin:0;
      }

      .card{
        width:100%;
        height:100%;
        border:none;
        border-radius:0;
      }

      .card-body{
        height:100%;
        display:flex;
```

```
flex-direction:column;
padding:10px;
overflow:hidden;
}

.dashboard-title{
text-align:center;
margin-bottom:10px;
flex-shrink:0;
}

.tableauPlaceholder{
flex:1;
width:100%;
overflow:hidden;
position:relative;
}
</style>
</head>

<body>

<div class="container-fluid">

<div class="card shadow">

<div class="card-body">

<h4 class="dashboard-title">
Regional Energy Analysis
</h4>

<div class="tableauPlaceholder"
id="viz1781513341115">

<noscript>
<a href="#">

</a>
```

```
</noscript>

<object class="tableauViz" style="display:none;">

    <param name="host_url"
    value="https%3A%2F%2Fpublic.tableau.com%2F"
    />

    <param name="embed_code_version"
    value="3" />

    <param name="path"
    value="shared/R4NS6ZMPM" />

    <param name="toolbar"
    value="yes" />

    <param name="static_image"
value="https://public.tableau.com/static/images/R4/R4NS6ZMPM/1.png" />

    <param name="animate_transition"
    value="yes" />

    <param name="display_static_image"
    value="yes" />

    <param name="display_spinner"
    value="yes" />

    <param name="display_overlay"
    value="yes" />

    <param name="display_count"
    value="yes" />

    <param name="language"
    value="en-US" />

    <param name="filter"
    value="publish=yes" />
```



DASHBOARD-3.html

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
```

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```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Dashboard 03</title>

<link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.
min.css" rel="stylesheet">

<style>
html, body {
height: 100%;
margin: 0;
overflow: hidden;
}

.container-fluid {
height: 100vh;
padding: 10px;
}

.card {
height: calc(100vh - 20px);
}

.card-body {
height: 100%;
display: flex;
flex-direction: column;
}

.tableauPlaceholder {
flex: 1;
width: 100%;
}
</style>
</head>

<body>

<div class="container-fluid">

<div class="card shadow h-100">
<div class="card-body">
```



```
<h4 class="text-center mb-3">
    Demand Forecasting
</h4>

<div class="tableauPlaceholder"
    id="viz1781513469591"
    style="position:relative;">

    <noscript>
        <a href="#">
            
        </a>
    </noscript>

    <object class="tableauViz" style="display:none;"

        <param name="host_url"
value="https%3A%2F%2Fpublic.tableau.com%2F"
        />

        <param name="embed_code_version"
            value="3" />

        <param name="site_root"
            value="" />

        <param name="name"
value="PluggingintotheFutureAnExplorationofElectricityConsumptionPatte
rnsDashboard-
3/PluggingintotheFutureAnExplorationofElectricityConsumptionPatterns-
Dashboard3" />

        <param name="tabs"
            value="no" />
```

```
<param name="toolbar"
      value="yes" />

<param name="static_image"
value="https://public.tableau.com/static/images/Pl/PluggingintotheFutu
reAnExplorationofElectricityConsumptionPatternsDashboard-
3/PluggingintotheFutureAnExplorationofElectricityConsumptionPatterns-
Dashboard3/1.png" />

<param name="animate_transition"
      value="yes" />

<param name="display_static_image"
      value="yes" />

<param name="display_spinner"
      value="yes" />

<param name="display_overlay"
      value="yes" />

<param name="display_count"
      value="yes" />

<param name="language"
      value="en-US" />

<param name="filter"
      value="publish=yes" />

</object>
</div>

</div>

</div>

<script>
```

```
var divElement = document.getElementById('viz1781513469591');
var vizElement = divElement.getElementsByTagName('object')[0];
```

```
    vizElement.style.width = '100%';
    vizElement.style.height = '100%';
```

```
    var scriptElement = document.createElement('script');
    scriptElement.src =
    'https://public.tableau.com/javascripts/api/viz_v1.js';
```

```
    vizElement.parentNode.insertBefore(
        scriptElement,
        vizElement
    );
```

```
    </script>
```

```
    </body>
```

```
    </html>
```

Story.html

```
    <!DOCTYPE html>
```

```
    <html lang="en">
```

```
        <head>
```

```
            <meta charset="UTF-8">
```

```
            <meta name="viewport" content="width=device-width, initial-
            scale=1.0">
```

```
            <title>Data Story</title>
```

```
            <!-- Bootstrap CSS -->
```

```
                <link
```

```
                href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.
                min.css" rel="stylesheet">
```

```
                <style>
```

```
                    body {
```

```
                        background: #f4f7fc;
```

```
                        font-family: Arial, sans-serif;
```

```
                    }
```

```
                    .container {
```

```
max-width: 1400px;
}

h1 {
  text-align: center;
  color: #0d6efd;
  font-weight: bold;
  margin-bottom: 30px;
}

.card {
  border: none;
  border-radius: 15px;
  overflow: hidden;
}

.card-body {
  text-align: center;
  padding: 30px;
}

h3, h4 {
  font-weight: 600;
  margin-bottom: 20px;
}

p {
  max-width: 900px;
  margin: 0 auto 15px;
  line-height: 1.8;
  color: #555;
  font-size: 1.05rem;
}

.tableauPlaceholder {
  width: 100%;
  margin: auto;
  display: flex;
  justify-content: center;
  align-items: center;
}
```

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```
.story-section {
  background: #fff;
  border-radius: 15px;
}

@media (max-width: 768px) {
  .tableauPlaceholder {
    height: 80vh !important;
  }
}

h1 {
  font-size: 2rem;
}

h3, h4 {
  font-size: 1.4rem;
}
}
</style>
</head>

<body>

<div class="container py-5">

<h1 class="display-6">Data Story</h1>

<!-- Tableau Story -->
<div class="card shadow-lg mb-5">
  <div class="card-body">

    <h4>
      Plugging into the Future: An Exploration of
      Electricity Consumption Patterns
    </h4>

    <div class='tableauPlaceholder'
      id='viz1781513605935'
      style='position:relative;width:100%;height:100vh;'>

      <noscript>
        <a href='#'>
```

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```
<img alt='Electricity Consumption Story'

src='https://public.tableau.com/static/images/Pl/PluggingintotheFuture
AnExplorationofElectricityConsumptionPatternsStory/Story1/1_rss.png'
style='border:none'>
</a>
</noscript>

<object class='tableauViz' style='display:none;'>

  <param name='host_url'
value='https%3A%2F%2Fpublic.tableau.com%2F'
/>

  <param name='embed_code_version'
value='3' />

  <param name='site_root'
value='' />

  <param name='name'
value='PluggingintotheFutureAnExplorationofElectricityConsumptionPatte
rnsStory/Story1' />

  <param name='tabs'
value='no' />

  <param name='toolbar'
value='yes' />

  <param name='static_image'
value='https://public.tableau.com/static/images/Pl/PluggingintotheFutu
reAnExplorationofElectricityConsumptionPatternsStory/Story1/1.png' />

  <param name='animate_transition'
value='yes' />

  <param name='display_static_image'
value='yes' />
```

```
<param name='display_spinner'
      value='yes' />

<param name='display_overlay'
      value='yes' />

<param name='display_count'
      value='yes' />

<param name='language'
      value='en-US' />

<param name='filter'
      value='publish=yes' />

</object>

</div>

</div>
</div>

<!-- Story Explanation -->
<div class="card shadow-lg story-section">
  <div class="card-body">

    <h3>The Story Behind Electricity Consumption</h3>

    <p>
      Electricity plays a vital role in modern society.
      As populations grow and industries expand, energy
      demand
      continues to rise.
    </p>

    <p>
      Through our analysis, we discovered key consumption
      patterns across different sectors and time periods.
      Seasonal changes and economic activities significantly
      influence electricity demand.
    </p>
```



```
<p>
Forecasting future energy needs can help governments,
industries, and utility providers improve energy
efficiency and sustainability planning.
</p>
```

```
<p>
This project highlights the importance of data-driven
decision making in building a smarter and greener
future.
</p>
```

```
</div>
</div>
</div>
<!-- Tableau Script -->
<script type='text/javascript'>

    var divElement =
document.getElementById('viz1781513605935');

    var vizElement =
divElement.getElementsByTagName('object')[0];

    vizElement.style.width = '100%';
    vizElement.style.height = '100vh';

    var scriptElement =
document.createElement('script');
    scriptElement.src =
'https://public.tableau.com/javascripts/api/viz_v1.js';

    vizElement.parentNode.insertBefore(
        scriptElement,
        vizElement
    );

</script>
```

```
</body>  
</html>
```

Style.css

```
body{  
  background-color:#f8f9fa;  
}  
  
.hero-section{  
  background: linear-gradient(  
    135deg,  
    #0d6efd,  
    #20c997  
  );  
  padding:120px 0;  
}  
  
.card{  
  border:none;  
  transition:0.3s;  
}  
  
.card:hover{  
  transform:translateY(-5px);  
}  
  
h1,h2,h3{  
  font-weight:700;  
}
```

